SAVINDU WANNIGAMA

Engineering Undergraduate

+94 77 625 9252

Savindu Wannigama

savinduwannigama

e17369@eng.pdn.ac.lk

EDUCATION

University of Peradeniya — ***November, 2018 - Present***

BSc. Engineering (Hons.) in Computer Engineering — *G.P.A: **4.0**/4.0 Rank: **2**/415*

Trinity College, Kandy — ***2008 - 2017***

G.C.E Advanced Level Examinations — *Grades: 3As Z-Score: 2.6383*

*National Rank: **6**/ 32075 Provincial Rank: **1**/2784*

PROJECTS

Sobriety Detection using Mobile Phone Gyroscope Data |

ML, DL, Python, TensorFlow, JavaScript, No-SQL, MQTT

Personal Contributions:

- Extracted practicable features from raw time series gyroscope readings using data segmentation and windowing, and used Principal Component Analysis (PCA), Feature Importance etc. to select the minimal subset of features for accurate performance of the ML models.
- Built an API to be used by mobile phones to post streams of real time gyroscopic readings, and get the classifier results.

Remote Proctoring System |

JavaScript, Node, No-SQL, Raspberry Pi, Linux, Jitsi-Meet

Personal Contributions:

- Designed and built a proctoring device based on a Raspberry Pi micro controller which acted as a bridge for a live video stream between all the students and a set of proctors. (Jitis Meet was used to integrate live video streaming into the web and desktop applications of the system.)
- Designed and built a system to store and handle exams and courses of a set of students registered in a system which implemented a scalable MongoDB database and and a Node based server running on a Linux server to serve requests from several users and store and auto update details about exams, courses, students etc

Reconstructing Highly Degraded License Plates Images |

Python, OpenCV

Personal Contributions:

- Demonstrated how traditional image processing techniques such as Otsu thresholding, morphological transformation, contouring, spatial and frequency domain filtering, and degradation modeling can be used to restore highly distorted images along with the limitations of the traditional image processing theoretical techniques.

8-Bit Single Cycle Processor |

Verilog-HDL

- Designed and prototyped an 8-bit single cycle CPU with associated memory hierarchy, which includes an arithmetic logic unit, register file, control logic, forwarding unit, data memory, data cache, instruction memory and instruction cache.

Applet to generate and display fractals |

Java, Multi-Threading

- Built a simple software to generate and display two fractal sets; Mandelbrot and Julia, according to user preferences.

ACHIEVEMENTS

IEEEXtreme 16.0 (International) — ***October, 2022***

A 24 hour algorithmic programming competition organized by IEEE

- National Rank - 34/311
- Global Rank - 768/6373

IEEEXtreme 14.0 (International) — ***October, 2020***

A 24 hour algorithmic programming competition organized by IEEE

- National Rank - 32/180
- Global Rank - 311/2155

Hacktitude (National) — ***January, 2020***

A 12 hour inter-university hackathon organized by 99x Technology (Pvt) Ltd.

- National Rank - 32/180 (out of 200+ teams)

Hackdown (National) — ***January, 2020***

A 3 hour inter-university coding competition organized by IEEE WIE

- Rank - 35 (Out of 200+ teams)

Dialog Merit Scholarship(National) — ***November, 2018 - Present***

Granted by Dialog Axiata PLC

- Recipient of a scholarship granted to the top performing student from each district in the Physical Sciences stream at the G.C.E. Advanced Level Examinations.

WORK EXPERIENCE

Teaching Assistant: CO322-2022: Data Structures and Algorithms — ***Sep, 2021 - Present***

Department of Computer Engineering, University of Peradeniya

- Supervise weekly lab sessions, create questions for tutorials.

Teaching Assistant: CO321-2022: Embedded Systems — ***Sep, 2021 - Present***

Department of Computer Engineering, University of Peradeniya

- Supervise and mark weekly lab sessions.

Teaching Assistant: CO224-2021: Computer Architecture — ***Nov, 2021 - Jan, 2022***

Department of Computer Engineering, University of Peradeniya

- Conducted tutorial sessions marking schemes for labs, supervised online lab session.
- Invigilated online quizzes every other week.

Teaching Assistant: CO222: Programming Methodology — ***May, 2021 - Sep, 2021***

Department of Computer Engineering, University of Peradeniya

- Supervised weekly lab sessions, created questions for online quizzes on the C programming Language.
- One-on-one sessions with students to tutor them on the C programming language and programming methodology concepts.

Member of the Web Consulation Team — ***Aug, 2021 - Present***

University of Peradeniya

- A team of engineering undergraduates volunteering to provide web consultation services to make the digital presence of the University of Peradeniya more impressive and optimized.
- The team is supervised by Prof. Roshan G. Ragel and backed by recent graduates of Department of Computer Engineering, University of Peradeniya.

RELEVANT COURSEWORK

- Programming Methodology
- Discrete Mathematics
- Probability and Statistics
- Ordinary Differential Equations
- Software Construction
- Database Systems
- Calculus II
- Numerical Methods
- Network and Web Application Design
- Data Structures and Algorithms
- Operating Systems
- Software Engineering
- Image Processing
- Machine Learning and Data Mining
- Signal Processing

TECHNICAL SKILLS

Languages:

C, Java, Python, JavaScript, SQL, NoSQL, Verilog HDL, ARM Assembly

Developer Tools:

Git, VS Code, Eclipse, Docker, Google Cloud Platform, Node

Technologies/ Frameworks:

Linux, GitHub, ExpressJS

LEADERSHIP/ EXTRA-CURRICULAR ACTIVITIES

Sports

Tennis

- Captain of the school Tennis team (2016/2017)
- Member of the university Tennis team (2019 – Present)
- University half-colors in Tennis (2019)

Other

Extra-curricular Activities

- Member of several clubs and societies in University and school including the Music Society and the Ceylon Dramatic Society of the University of Peradeniya, and the Cadet Band and the Science Society of Trinity College Kandy.

REFERENCES

Prof. Roshan Ragel
Head of Department,
Department of Computer Engineering,
Faculty of Engineering,
University of Peradeniya, Sri Lanka.
roshanr@eng.pdn.ac.lk

Dr. Isuru Nawinne
Senior Lecturer,
Department of Computer Engineering,
Faculty of Engineering,
University of Peradeniya, Sri Lanka
isurunawinne@eng.pdn.ac.lk